

Skills Summary

Integrating Rational Functions With Partial Fractions

Use this reference while completing your worksheet or when studying.

The whole method in one line: a rational function with distinct linear factors downstairs equals a sum of constants over those factors, and each piece integrates to a logarithm — $\int \frac{dx}{x-r} = \ln|x-r| + C$.

- **Check the degree first.** If the numerator's degree is not lower than the denominator's, do polynomial division before splitting.
- **Find the constants by substitution.** Clear denominators, then put x equal to each root in turn: every term but one vanishes.
- **Keep the absolute values** and combine differences of logarithms into one logarithm at the end.

1. Splitting into distinct linear factors

Write $\frac{N(x)}{(x-r_1)(x-r_2)} = \frac{A}{x-r_1} + \frac{B}{x-r_2}$, clear denominators, and substitute $x = r_1$ then $x = r_2$. Check the split by recombining over the common denominator — faster than differentiating.

Example: Find $\int \frac{4}{(x-1)(x+3)} dx$.

Step 1. The factors are distinct and linear and the numerator's degree is lower, so constants over each factor are all that is needed.

Step 2. $4 = A(x+3) + B(x-1)$: at $x = 1$, $4 = 4A$ so $A = 1$; at $x = -3$, $4 = -4B$ so $B = -1$. Integrating each piece gives the logarithms.

$$\ln|x-1| - \ln|x+3| + C$$

Try it: Find $\int \frac{6}{(x-2)(x+1)} dx$.
Check: $\ln|x+1| - 2\ln|x-2| + C$

2. Definite integrals after the split

Split first, antidifferentiate, then substitute the limits. Combine the logarithms into one before evaluating: it turns four log values into a single quotient and removes most of the arithmetic.

Example: Evaluate $\int_1^2 \frac{3}{x(x+3)} dx$.

Step 1. A definite integral changes nothing about the algebra — decompose exactly as before, and save the limits for the last line.

Step 2. $3 = A(x+3) + Bx$ gives $A = 1$, $B = -1$, so the antiderivative is $\ln \left| \frac{x}{x+3} \right|$, and evaluating gives $\ln \frac{2}{5} - \ln \frac{1}{4}$.

$$\ln \frac{8}{5}$$

Try it: Evaluate $\int_0^1 \frac{2}{(x+1)(x+3)} dx$.
check: $\ln \frac{2}{3}$

3. Improper integrals: split, then take a limit

$\int_a^\infty f = \lim_{b \rightarrow \infty} \int_a^b f$. Decompose, evaluate on $[a, b]$, **combine into a single logarithm**, and only then let $b \rightarrow \infty$. A quotient like $\frac{b}{b+1}$ tends to 1, so its logarithm tends to 0; separate logarithms would give $\infty - \infty$ and decide nothing.

Example: Evaluate $\int_1^\infty \frac{1}{x(x+1)} dx$.

Step 1. An infinite limit is not a number to substitute, so the integral is defined as the limit of the proper integral on $[1, b]$.

Step 2. $\frac{1}{x} - \frac{1}{x+1}$ integrates to $\ln \left| \frac{x}{x+1} \right|$; on $[1, b]$ that is $\ln \frac{b}{b+1} + \ln 2$, and the first term tends to $\ln 1 = 0$.

$$\ln 2$$

Try it: Evaluate $\int_2^\infty \frac{4}{x(x+2)} dx$.
check: $\ln \frac{1}{4}$

Watch out. A coefficient belongs outside its logarithm: $2 \ln |x-2|$ is $\ln(x-2)^2$, not $\ln |2x-4|$. And decaying to zero does not make an improper integral converge — $\int_0^\infty \frac{dt}{t+1}$ diverges even though the integrand tends to 0. What matters is how fast.